Lecture 5: Sections 1.1, 1.2, and 1.3

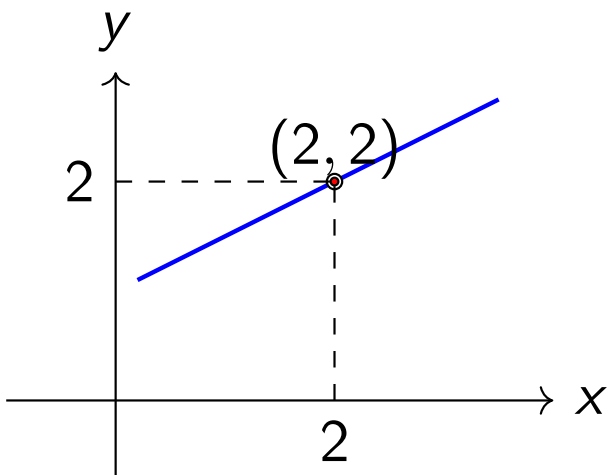
Limit

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The Basic Idea of Limits

The limit describes the behavior of a function as the input approaches some value.



As x approaches 2, $f(x)$ approaches 2

$$\lim_{x \to a} f(x) = L$$

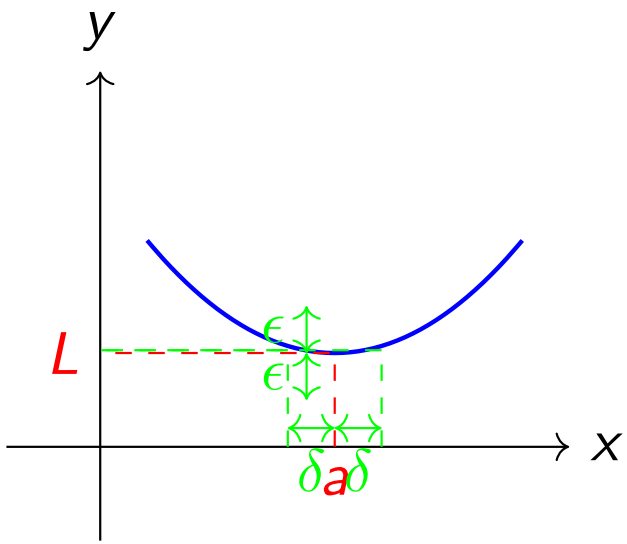
"the limit of f(x) as x approaches a equals L"

Formal Definition of Limit

ϵ-δ Definition

$$\lim_{x \to a} f(x) = L$$

means: For every $\epsilon > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.



One-Sided Limits

Left-Hand Limit

$$\lim_{x \to a^-} f(x) = L$$

Approaching from the left side only

Right-Hand Limit

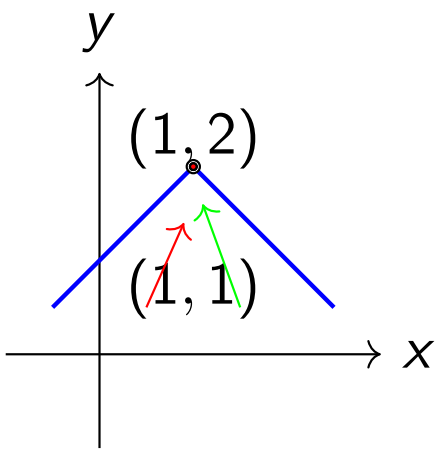
$$\lim_{x \to a^+} f(x) = L$$

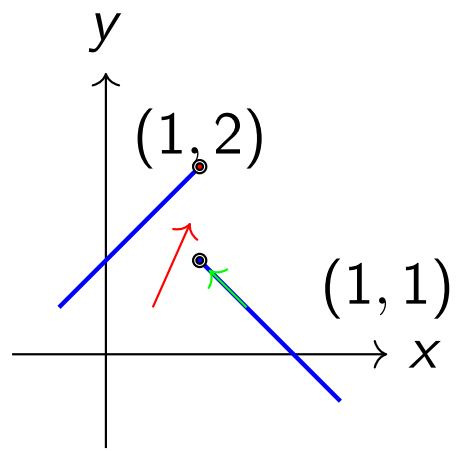
Approaching from the right side only

Key Theorem

$$\lim_{x \to a} f(x) = L \quad \text{if and only if} \quad \lim_{x \to a^-} f(x) = L \quad \text{and} \quad \lim_{x \to a^+} f(x) = L$$

One-Sided Limits: Graphical Examples





$\lim_{x \to 1^-} f(x) = 2$, $\lim_{x \to 1^+} f(x) = 1$

$\lim_{x \to 1^-} f(x) = 2$, $\lim_{x \to 1^+} f(x) = 2$

Properties of Limits

Basic Properties

Assume $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist:

- $\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$
- $\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$
- $\lim_{x\to a}[cf(x)] = c\lim_{x\to a} f(x)$
- $\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$
- $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$, if $\lim_{x\to a} g(x) \neq 0$
- $\lim_{x\to a}[f(x)]^n = [\lim_{x\to a} f(x)]^n$

Special Limits

- $\lim_{x\to a} c = c$ (constant function)
- $\lim_{x\to a} x = a$
- $\lim_{x\to a} x^n = a^n$

Limits of Polynomial and Rational Functions

Direct Substitution

For polynomials and rational functions (where denominator $\neq 0$):

$$\lim_{x \to a} f(x) = f(a)$$

Example

Evaluate using direct substitution:

1. $\lim_{x \to 2}(3x^2 - 2x + 1)$
2. $\lim_{x \to -1} \frac{x^2 - 1}{x + 2}$
3. $\lim_{x \to 3} \sqrt{x + 1}$
4. $\lim_{x \to 0} \frac{\sin x}{x}$ (special case)

Step-by-Step: $\lim_{x\to 2}(3x^2 - 2x + 1)$

$$3(2)^2 - 2(2) + 1 = 12 - 4 + 1 = 9$$

Limits with Indeterminate Forms

Common Indeterminate Forms

- $\frac{0}{0}$ (most common)
- $\frac{\infty}{\infty}$
- $0 \cdot \infty$
- $\infty - \infty$
- 0^0, 1^∞, ∞^0

Techniques for $\frac{0}{0}$ Forms

- Factoring
- Rationalizing
- Simplifying complex fractions

Example

Evaluate:

1. $\lim_{x \to 2} \frac{x^2-4}{x-2}$
2. $\lim_{x \to 9} \frac{\sqrt{x}-3}{x-9}$
3. $\lim_{x \to 1} \frac{x^3-1}{x-1}$

Step-by-Step: Indeterminate Forms

Example 1: $\lim_{x\to 2} \frac{x^2-4}{x-2}$

$$\begin{aligned}\lim_{x\to 2} \frac{x^2-4}{x-2} &= \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} \\ &= \lim_{x\to 2} (x+2) \\ &= 2+2 = 4\end{aligned}$$

Example 2: $\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}$

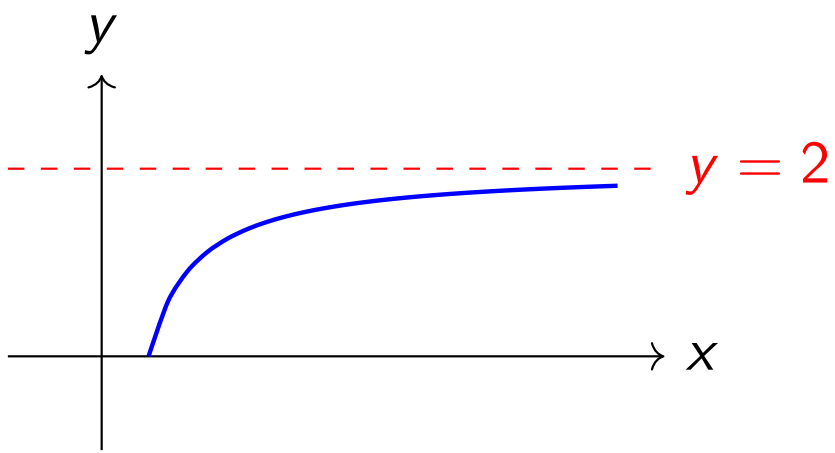
$$\begin{aligned}\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9} &= \lim_{x\to 9} \frac{\sqrt{x}-3}{x-9} \cdot \frac{\sqrt{x}+3}{\sqrt{x}+3} \\ &= \lim_{x\to 9} \frac{x-9}{(x-9)(\sqrt{x}+3)} \\ &= \lim_{x\to 9} \frac{1}{\sqrt{x}+3} = \frac{1}{6}\end{aligned}$$

Limits at Infinity

Definition

$$\lim_{x \to \infty} f(x) = L$$

means $f(x)$ approaches L as x increases without bound.



As $x \to \infty$, $f(x) \to 2$

Example

Evaluate:

1. $\lim_{x\to\infty} \frac{3x^2-2x+1}{2x^2+x-5}$
2. $\lim_{x\to\infty} \frac{2x+1}{\sqrt{x^2+3}}$
3. $\lim_{x\to-\infty} e^x$

Infinite Limits

Definition

$$\lim_{x \to a} f(x) = \infty$$

means $f(x)$ increases without bound as x approaches a.

Example

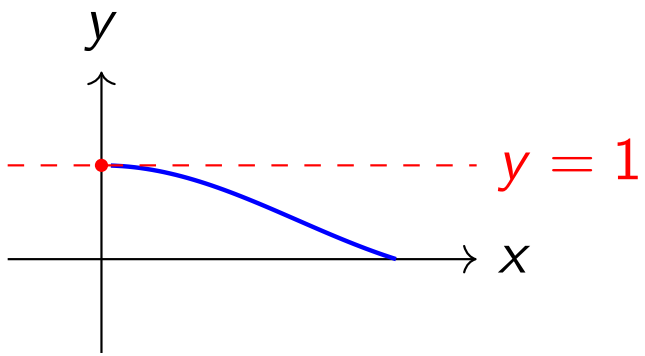
Evaluate:

1. $\lim_{x \to 0^+} \frac{1}{x}$
2. $\lim_{x \to 2^-} \frac{1}{x-2}$
3. $\lim_{x \to 0} \frac{1}{x^2}$

Special Trigonometric Limits

Important Trigonometric Limits

- $\lim_{x\to 0} \frac{\sin x}{x} = 1$
- $\lim_{x\to 0} \frac{1-\cos x}{x} = 0$
- $\lim_{x\to 0} \frac{\tan x}{x} = 1$



$\lim_{x\to 0} \frac{\sin x}{x} = 1$

Example

Evaluate:

1. $\lim_{x \to 0} \frac{\sin(3x)}{x}$
2. $\lim_{x \to 0} \frac{\tan(2x)}{x}$
3. $\lim_{x \to 0} \frac{1-\cos x}{x^2}$

Piecewise Functions and Limits

Piecewise Function Limit

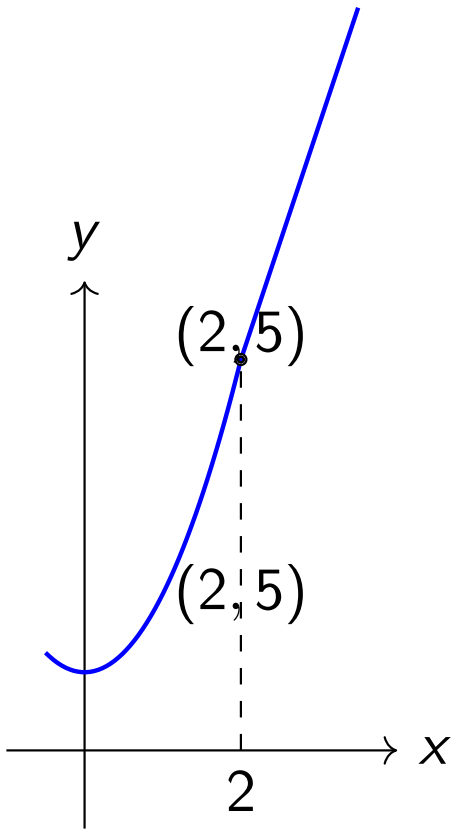
Evaluate left and right limits separately at transition points.

Example

Let $f(x) = \begin{cases} x^2 + 1 & \text{if } x < 2 \\ 3x - 1 & \text{if } x \geq 2 \end{cases}$

Find:

1. $\lim_{x \to 2^-} f(x)$
2. $\lim_{x \to 2^+} f(x)$
3. $\lim_{x \to 2} f(x)$


y
(2, 5)
(2, 5)
x
2

Practice Problems: Basic Limits

Direct Evaluation

Evaluate:

1. $\lim_{x \to 3}(2x^2 - 5x + 1)$
2. $\lim_{x \to -1} \frac{x^3+1}{x+1}$
3. $\lim_{x \to 4} \sqrt{x + 5}$
4. $\lim_{x \to 0} \frac{\sin(5x)}{x}$
5. $\lim_{x \to 2} \frac{x^2-4}{x-2}$

One-Sided Limits

Given $f(x) = \begin{cases} x+2 & \text{if } x < 1 \\ x^2 & \text{if } x \geq 1 \end{cases}$, find:

1. $\lim_{x \to 1^-} f(x)$
2. $\lim_{x \to 1^+} f(x)$
3. $\lim_{x \to 1} f(x)$

Practice Problems: Intermediate Level

Indeterminate Forms

Evaluate:

1. $\lim_{x\to 1} \frac{x^2-3x+2}{x-1}$
2. $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$
3. $\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$
4. $\lim_{x\to 2} \frac{x^3-8}{x-2}$
5. $\lim_{x\to 1} \frac{x^4-1}{x-1}$

Limits at Infinity

Evaluate:

1. $\lim_{x\to\infty} \frac{3x^2-2x+1}{x^2+4}$
2. $\lim_{x\to-\infty} \frac{2x+1}{\sqrt{x^2+1}}$
3. $\lim_{x\to\infty} \frac{4x^3-x}{2x^3+5}$

Practice Problems: Advanced Level

Challenging Limits

Evaluate:

1. $\lim_{x \to 0} \frac{\sin(3x)}{\tan(2x)}$
2. $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$
3. $\lim_{x \to 0} \frac{e^x - 1}{x}$
4. $\lim_{x \to 1} \frac{x^{1/3} - 1}{x^{1/2} - 1}$
5. $\lim_{x \to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

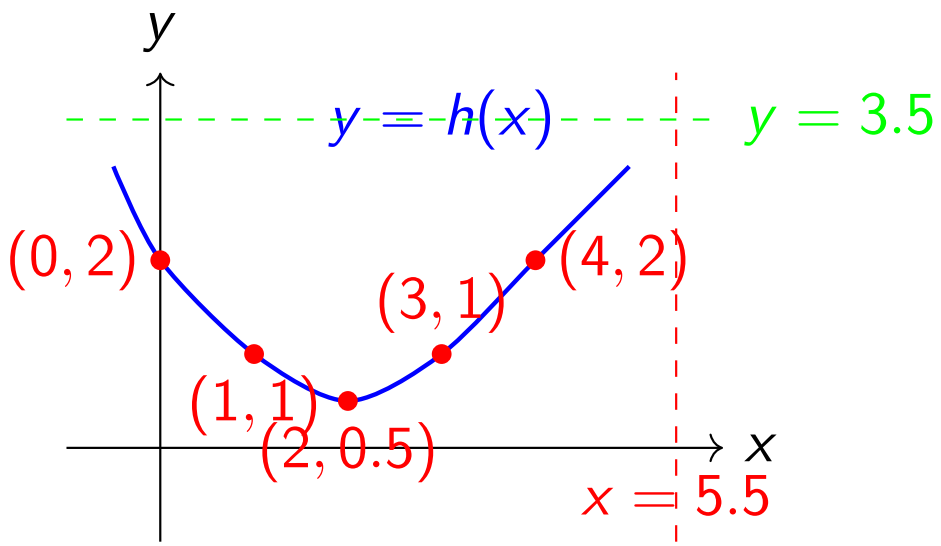
Piecewise Function Analysis

Let $g(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Find:

1. $\lim_{x \to 0^-} g(x)$
2. $\lim_{x \to 0^+} g(x)$
3. $\lim_{x \to 0} g(x)$

Graphical Limits Practice



1. $\lim_{x \to 1} h(x)$
2. $\lim_{x \to 2} h(x)$
3. $\lim_{x \to 3} h(x)$
4. $\lim_{x \to 5.5^-} h(x)$
5. $\lim_{x \to 5.5^+} h(x)$
6. $\lim_{x \to \infty} h(x)$
7. $\lim_{x \to -\infty} h(x)$

Key Concepts and Summary

Fundamental Limit Concepts

- Limit describes function behavior near a point
- One-sided limits approach from left/right
- Limit exists only if left and right limits equal
- Direct substitution works for continuous functions

Important Techniques

- Factoring for $\frac{0}{0}$ forms
- Rationalizing for radical expressions
- Dividing by highest power for limits at infinity
- Special trigonometric limits

Applications

- Foundation for derivatives
- Instantaneous rates of change
- Continuity analysis
- Asymptotic behavior
- Real-world modeling (physics, engineering, economics)